DTMES 2026 · AFRICA ELECTRONICS INNOVATION CHALLENGE · BUILT STAGE

RFID Pay

Tap to pay your matatu fare. M-Pesa, no cash, no conning.

A low-cost ESP32 + RFID fare system. Passengers tap a card or wristband, pay the fare with M-Pesa, and the owner sees every shilling live.

ESP32 + RC522 | M-Pesa Daraja | WhatsApp + USSD



Team HYDRA-PAY

Technical University of Kenya

THE PROBLEM

Matatu cash fares leak money and invite conning



Cash leaks

Fares are collected in cash with no record. Owners lose part of the daily takings and can't prove how much was collected.



Conning & fake M-Pesa

Fake confirmation SMSs, money sent to the wrong number, overcharging and fights over change.



Slow & unsafe

Counting coins and giving change at every stop slows boarding, and crews carrying cash are targets for theft.

Every passenger has M-Pesa. The fare still ends up in a pocket.

OUR SOLUTION

One tap pays the fare straight to the owner



Tap on boarding

A reader by the door reads the passenger's card or wristband instantly.

2

M-Pesa prompt

The fare is sent as a Daraja STK push to the passenger's phone, paid to the owner's till.

3

Enter PIN

The passenger approves on their own phone. No cash, no change, no conductor pocket.

4

Receipt

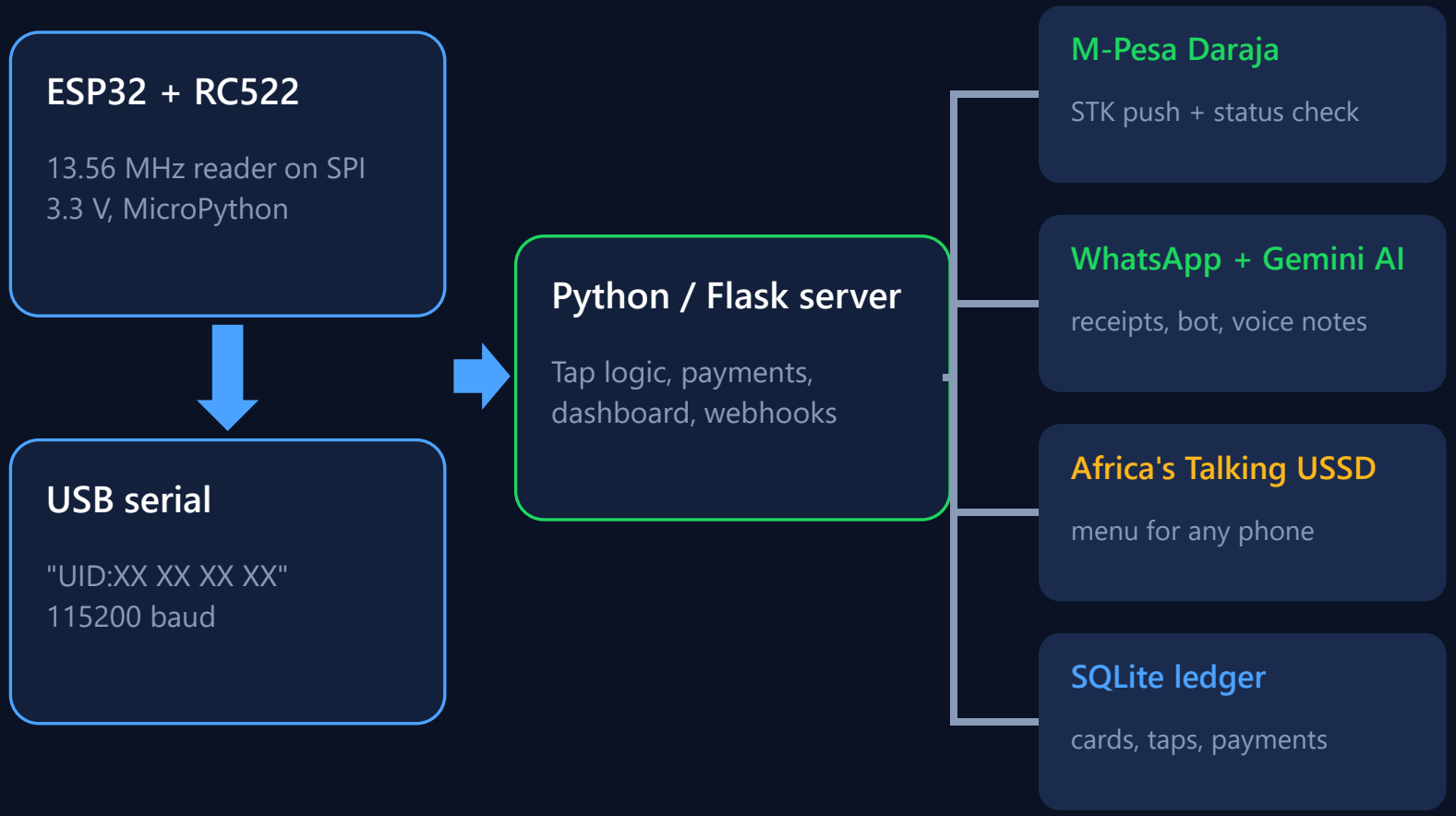
WhatsApp receipt with the M-Pesa code. The owner's dashboard updates live.

Pay-per-tap:
Every tap IN sends one M-Pesa prompt for the fare. Tap OUT is free.

Commuter wallet:
Regulars top up once. Each tap deducts the fare instantly, no PIN wait.

TECHNICAL DESIGN & SOUNDNESS

How it is built



Engineering decisions

Reliable reads: firmware debounce, wake-up of half-read cards, 3× retry and max antenna gain (48 dB).
Money is safe: every payment is confirmed with Safaricom twice (callback + status query) and recorded exactly once under a lock.
Auditable: append-only ledger; undo is a reversal entry, nothing is deleted. CSV export.
Secure admin: admin PIN for cards, cash and undo; remote login is blocked while the PIN is default.
Self-healing: auto-detects the USB port and reconnects if the reader is unplugged.

FUNCTIONALITY / PROOF OF WORK

It works today, and you can try it at our table

26
M-Pesa prompts sent through Daraja in testing

9
payments completed with a PIN on a real phone

2
payment modes: pay-per-tap & wallet

Daraja sandbox test runs, 22–23 Oct 2026

Live demo

Tap a card like boarding a matatu: watch the prompt arrive, the dashboard update and the WhatsApp receipt land.

Working features

- ✓ Card and key-tag taps read and logged as IN / OUT
- ✓ STK push sent to the card owner's phone for the card's price
- ✓ WhatsApp receipt with the M-Pesa code after payment
- ✓ WhatsApp bot: card, payments, one-message number change
- ✓ AI assistant in English, Kiswahili and Sheng, plus voice notes
- ✓ Live dashboard: register cards, daily totals, CSV export
- ✓ Lost card: move the account to a new card in one click
- ✓ USSD menu on any phone via Africa's Talking

NOVELTY & INNOVATION

What makes RFID Pay different

Can't be conned



A fare only shows as PAID after Safaricom confirms it twice (callback + status query). Fake SMSs and screenshots don't count.

Owner sees every fare



Every tap, payment and M-Pesa code appears live on the dashboard, with daily totals and CSV. Nothing to skim.

Low-cost hardware



Off-the-shelf ESP32 + RC522 by the door. Works with 13.56 MHz cards, key tags and wristbands. No costly POS terminal.

Any phone, any change



USSD on basic phones. New number? Send it on WhatsApp. Lost card? Tap a new one and move the account. No re-registration.

IMPACT & PRACTICAL RELEVANCE

Who benefits

Passengers

- No cash or change needed
- A WhatsApp receipt for every trip
- Regulars top up once and just tap

Owners & SACCOs

- Every fare reaches the owner's till
- Live takings per vehicle, daily CSV
- No skimming, no fake payments

Crews & the public

- No cash on board to rob
- Faster boarding at every stage
- Fewer fare disputes

The same system fits school buses, canteens, events and estates.

WHAT'S NEXT

Roadmap: live USSD, offline reader, multi-currency & crypto

NEXT

Live USSD short code

Our USSD menu runs on Africa's Talking today. Next: a live short code any passenger can dial.

NEXT

Offline reader

The reader stores taps when the network drops on the road and syncs when it is back.

PILOT

SACCO pilot

Readers in a few matatus on one route, with a fleet dashboard for the SACCO.

LATER

Multi-currency & crypto

Pay in KES, USD, UGX or TZS, or in crypto (BTC / Lightning, USDT). The owner always receives KES.

Already built: RFID + M-Pesa + WhatsApp + USSD + dashboard. Next: every matatu, every route, even offline.

INDUSTRY ENGAGEMENT

Meeting Safaricom leadership



With Safaricom's former Chief Financial Services Officer



With Safaricom's Chief Technology & Information Officer

Tap. Pay. Done.

RFID Pay: cashless matatu fares on M-Pesa, with receipts on WhatsApp.

Questions? Try the live demo

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